

HOSPITAL INFORMATION SYSTEM

User Documentation

Complete Guide for Staff and Administrators

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TABLE OF CONTENTS

1 Getting Started

- 1.1 System Requirements
- 1.2 Launching the Application
- 1.3 Logging In
- 1.4 User Roles Overview

2 Navigation & Interface

- 2.1 Main Window Layout
- 2.2 Sidebar Navigation
- 2.3 Dashboard

3 Patient Registration

- 3.1 Auto-Assigned Patient ID
- 3.2 Manual Patient ID (Admin Only)
- 3.3 Registration Fields Reference
- 3.4 Duplicate Patient Detection

4 Search & Patient Records

- 4.1 Searching for Patients
- 4.2 Editing a Patient Record
- 4.3 Deleting a Patient Record

5 User Management (Admin Only)

- 5.1 Adding a New User
- 5.2 Editing & Deleting Users

6 Field Reference

7 Troubleshooting

1 GETTING STARTED

1.1 System Requirements

Component	Requirement
Operating System	Windows 10 / 11 (64-bit)
Python	Python 3.10 or higher
Database	MySQL 8.0 or higher
Network	Local network access to database server (192.168.1.88)
Display	Minimum 1280 × 720 resolution
Dependencies	mysql-connector-python, tkcalendar (optional)

1.2 Launching the Application

- 1

Open a terminal or command prompt
Navigate to the application folder on your desktop or installation directory.
- 2

Run the application
Execute: `python main.py` — or double-click the provided shortcut.
- 3

Wait for startup
The system connects to the MySQL database. A login window will appear within a few seconds.

1.3 Logging In

The login screen is the first thing you see when the application starts. Enter your username and password, then press Enter or click **SIGN IN**.

■ Default administrator credentials on first run: username: admin | password: admin123 — Change the password immediately after first login.

■ If login fails, verify that the MySQL database server is running and reachable at the configured IP address.

1.4 User Roles Overview

The system uses two access levels. Your role determines which pages and actions are available to you.

Feature	Staff	Administrator
Dashboard	✓	✓
Patient Registration (auto ID)	✓	✓
Search & Edit Patients	✓	✓
Manual Patient Registration (set ID)	✗	✓
Add System Users	✗	✓
Manage / Delete Users	✗	✓

2 NAVIGATION & INTERFACE

2.1 Main Window Layout

After a successful login the main application window opens. It is divided into three fixed zones:

Zone	Location	Purpose
Top Bar	Top edge, full width	Shows the system name, logged-in username, and role badge.
Sidebar	Left, fixed 228 px wide	Navigation menu. All pages are accessed from here.
Content Area	Right, fills remaining	Displays the active page (dashboard, forms, tables, etc.).

2.2 Sidebar Navigation

Click any item in the dark sidebar to switch pages. The active page is highlighted in blue. Administrator accounts see additional items under the **ADMINISTRATION** section.

Menu Item	Role	Description
Dashboard	All	Overview statistics and recent patients.
Patient Registration	All	Register a new patient with an auto-assigned ID.
Search Patient	All	Search, view, edit, or delete patient records.
Manual Registration	Admin	Register a patient with a custom, manually typed ID.
User Registration	Admin	Create a new staff or administrator account.
User Management	Admin	View, edit roles, reset passwords, or delete users.
Exit Application	All	Close the application safely.

2.3 Dashboard

The dashboard is the home screen shown immediately after login. It provides a quick summary of system activity:

- **Total Patients** — total number of registered patient records.
- **System Users** (Admin) or **Your Role** (Staff) — account information.
- **Recent Registrations** — count of the 10 most recently added patients.
- **Recent Patients Table** — scrollable list of the latest 10 patient entries.

3 PATIENT REGISTRATION

3.1 Auto-Assigned Patient ID (Staff & Admin)

Navigate to **Patient Registration** in the sidebar. A patient ID is automatically generated and displayed at the top of the form, starting from **30000** and incrementing by 1 for each new patient. You cannot modify this ID — it is assigned by the system.

- 1

Fill in patient details
Complete all required fields. All text is saved in uppercase automatically.
- 2

Check for duplicates
Click Register Patient. If a patient with the same name and birth date exists, a warning appears.
- 3

Save the record
Click Register Patient to save. A confirmation dialog shows the assigned ID.
- 4

Form resets
After saving, the form clears and a new patient ID is pre-loaded, ready for the next registration.

3.2 Manual Patient Registration (Admin Only)

■ This page is only accessible to Administrator accounts.

Navigate to **Manual Registration** under the Administration section. This page lets an administrator enter any custom patient ID number. Use this when migrating records from another system or correcting ID assignments.

- 1

Enter the desired Patient ID
Type the ID in the Patient ID field at the top. A live indicator shows ✓ ID is available or ✗ ID already in use.
- 2

Complete patient details
Fill in all fields as normal. The Nationality field defaults to FILIPINO.
- 3

Save the record
Click Register Patient. The system blocks saving if the ID is already taken.
- 4

Save Anyway
If a duplicate name/birth date is detected, use Save Anyway after confirming you intend to create the record.

3.3 Registration Fields Reference

Field	Description	Required
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First Name	Patient's given name. Saved in uppercase.	Required
Middle Name	Patient's middle name.	Optional
Last Name	Patient's family name. Saved in uppercase.	Required
Gender	MALE, FEMALE, or OTHER. Select from dropdown.	Required
Civil Status	SINGLE, MARRIED, WIDOWED, or DIVORCED.	Required
Nationality	Defaults to FILIPINO. Edit if different.	Optional
Birth Date	Format: MM-DD-YYYY (e.g. 01-15-1990).	Optional
Birth Place	City or municipality of birth.	Optional
Phone Number	Contact phone number.	Optional
Email Address	Contact email address. Validated if entered.	Optional
Barangay	Barangay of current address.	Optional
Municipality	Municipality or city of current address.	Optional
Province	Province of current address.	Optional

3.4 Duplicate Patient Detection

Before saving, the system checks whether a patient with the same **First Name**, **Last Name**, and **Birth Date** already exists. If a match is found:

- A warning dialog appears explaining the potential duplicate.
- Click **Register Patient** again to cancel and review the details.
- Click **Save Anyway** to proceed and save the record regardless.

4 SEARCH & PATIENT RECORDS

4.1 Searching for Patients

Navigate to **Search Patient**. The search panel at the top provides three filter fields. You may use one or all three together:

Filter Field	Matches Against
ID or Full Name	Patient ID, first name, middle name, last name, or any combination.
First Name	Patients whose first name contains the entered text.
Last Name	Patients whose last name contains the entered text.

✓ Searches are case-insensitive. Press Enter in any search field to trigger the search instantly.

Results appear in the table below the search panel. Click any row to load that patient's details into the edit panel at the bottom of the page.

4.2 Editing a Patient Record

1 Search for the patient

Use the search panel to find the patient. Click their row in the results table.

2 Click Update Record

The edit panel below shows the patient's current data. Click Update Record to open the edit popup.

3 Edit the popup fields

All fields are pre-filled with existing data. Change only what needs updating.

4 Save Changes

Click Save Changes. A success message confirms the update. The results table refreshes automatically.

■ The Patient ID itself cannot be changed through the edit popup. To reassign an ID, delete the record and re-register using Manual Re-

4.3 Deleting a Patient Record

Select a patient from the results table, then click **Delete Record** in the edit panel. A confirmation dialog appears — click **Yes** to permanently remove the record.

- Deletion is permanent and cannot be undone. Confirm the patient ID before proceeding.

5 USER MANAGEMENT (ADMIN ONLY)

■ All pages in this section are accessible to Administrator accounts only.

5.1 Adding a New User

Navigate to **User Registration** in the sidebar.

Field	Description	Required
Username	Unique login name. 3–20 characters, letters/numbers/underscore only.	Required
Password	Minimum 6 characters.	Required
Confirm Password	Must match the Password field exactly.	Required
Role	Staff — standard access. Administrator — full access.	Required

1

Fill in all fields

Username, password, confirm password, and role are all required.

2

Click Register User

The system checks for duplicate usernames before saving.

3

Confirmation

A green success message appears and the form resets for the next user.

5.2 Editing & Deleting Users

Navigate to **User Management** in the sidebar.

The left panel lists all system accounts. Click any row to load the user's details on the right.

Changing a user's role:

1

Select the user

Click the user's row in the table on the left.

2

Change the Role

Use the Role dropdown on the right panel to select the new role.

3 Click Update User

The change is saved immediately. The table refreshes.

Resetting a password:**1 Select the user**

Click the user's row in the table.

2 Enter new password

Fill in New Password and Confirm Password fields.

3 Click Update User

The new hashed password is saved. Leave both fields blank to update only the role.

Deleting a user account:

Select a user and click **Delete User**. Confirm the dialog. The account is removed immediately. You cannot delete the account you are currently logged in as.

6 QUICK FIELD REFERENCE

This section summarises all form fields across the application for quick reference.

Patient Record Fields

Field	Description	Required
Patient ID	Unique identifier. Auto-generated (30000+) or manually assigned by Admin.	System
First Name	Patient's given name. Stored in uppercase.	Required
Middle Name	Patient's middle name. Stored in uppercase.	Optional
Last Name	Patient's family name. Stored in uppercase.	Required
Gender	MALE / FEMALE / OTHER.	Required
Civil Status	SINGLE / MARRIED / WIDOWED / DIVORCED.	Required
Nationality	Defaults to FILIPINO.	Optional
Birth Date	MM-DD-YYYY format (e.g. 03-22-1985).	Optional
Birth Place	City or municipality of birth.	Optional
Phone	Contact phone number.	Optional
Email	Contact email. Validated on entry.	Optional
Barangay	Barangay portion of current address.	Optional
Municipality	Municipality or city of current address.	Optional
Province	Province of current address.	Optional
Medical History	Free-text field for relevant medical notes.	Optional
Registered By	Auto-filled with the logged-in username.	System
Registration Date	Auto-filled with the current date and time.	System

User Account Fields

Field	Description	Required
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Username	Unique login name. 3–20 alphanumeric characters or underscore.	Required
Password	SHA-256 hashed. Minimum 6 characters at input.	Required
Role	Staff or Administrator. Controls page access.	Required
Created Date	Auto-filled when the account is created.	System
Last Login	Auto-updated on each successful login.	System

7 TROUBLESHOOTING

Problem: Cannot log in — 'Invalid username or password'

- Verify caps lock is off. Usernames and passwords are case-sensitive.
- Ask your Administrator to verify the account exists in User Management.
- Default admin credentials are: admin / admin123 (change after first use).

Problem: Application does not start — database connection error

- Confirm the MySQL server is running on the host machine.
- Check that the database IP (192.168.1.88) is reachable from your machine.
- Verify the database credentials in models/database.py or via environment variables DB_HOST, DB_USER, DB_PASSWORD, DB_NAME.

Problem: Patient registration fails silently

- Check the terminal/console for a printed error message.
- Ensure all required fields (First Name, Last Name) are filled in.
- If the Patient ID is already in use (Manual Registration), choose a different ID.

Problem: Search returns no results

- Confirm at least one search field is filled in before clicking Search.
- Try a shorter or partial search term.
- Verify the patient exists using a different field (e.g. search by ID instead of name).

Problem: Birth date not saving correctly

- The birth date field requires MM-DD-YYYY format (e.g. 01-15-1990).
- Do not use YYYY-MM-DD or other formats — the entry will not be stored.

Problem: Changes not reflected in the table after update

- Click the Refresh button (■) if available, or re-run the search.
- The dashboard refreshes on next navigation — click Dashboard in the sidebar.

For technical support or to report issues, contact your system administrator.